scPTR: Decomposing Post-Transcriptional Regulation at Single-Cell Resolution

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Abstract

Transcript abundance is jointly determined by mRNA synthesis and degradation, yet single-cell computational frameworks have addressed only the synthesis side. RNA velocity recovers transcriptional dynamics from spliced/unspliced ratios, but mRNA degradation—orchestrated by RNA-binding proteins (RBPs), microRNAs, and *cis*-regulatory elements—remains an unobserved axis. We present **scPTR**, a framework that estimates per-cell, per-gene mRNA degradation rates (γ) from standard scRNA-seq, discovers cell subpopulations defined by post-transcriptional programs, and infers RBP regulatory networks at single-cell resolution. scPTR's γ estimates correlate with published mRNA half-lives (Spearman $\rho{=}-0.81$ on sci-fate metabolic-labeling data), outperforming scVelo steady-state ($\rho{=}-0.37$) and velVI ($\rho{=}-0.28$); 59% of 215 miRNA target families are significantly enriched ($p{=}4.7{\times}10^{-65}$). Clustering in γ-space reveals *expression-invisible* post-transcriptional states in 3 of 8 pancreatic and 6 of 11 hippocampal cell types. Post-transcriptional velocity is orthogonal to RNA velocity and shows that degradation-rate changes *precede* expression changes for 54% of transition genes. Library-size-corrected RBP–target inference recovers known cancer dependencies. We additionally introduce DEEPPTR, a structured VAE with a kinetic decoder that factorizes the latent space into transcriptional and post-transcriptional components under a negative binomial observation model. Code, data, and notebooks reproducing every figure are released anonymously with the supplementary material.

1 Introduction

Steady-state mRNA abundance is set by the balance of synthesis and degradation. Single-cell computational biology has invested heavily in the synthesis side: RNA velocity [La Manno et al., 2018, Bergen et al., 2020] recovers transcriptional dynamics from the ratio of nascent (unspliced) to mature (spliced) reads, and modern deep-generative variants [Gayoso et al., 2024, Gao et al., 2022] now infer latent dynamical states under expressive priors. Degradation has remained out of view, despite its established role in setting cell identity and fate [Hentze et al., 2018]. RNA-binding proteins (RBPs), microRNAs, and $3'$ UTR *cis*-elements jointly determine transcript half-lives that vary by more than three orders of magnitude across the transcriptome and shift sharply during cell-state transitions, but no method to date estimates per-cell degradation rates from standard scRNA-seq.

This omission is consequential for three reasons. First, two cells with identical transcript abundances can occupy radically different post-transcriptional regimes—one stabilising short-half-life regulators, the other actively degrading them—and these *expression-invisible* states are precisely where regulatory commitment occurs. Second, the most direct biophysical handle on RBP function is the per-cell stability of its target transcripts, but in the absence of single-cell γ estimates, RBP networks must be reconstructed from co-expression alone, which conflates synthesis and decay. Third, in disease contexts—particularly cancer, where oncogenic stabilisation of short-lived transcripts is a known

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mechanism [Dempster et al., 2021]—the relevant signal is degradation, not abundance. A framework that read out per-cell degradation rates would *(i)* reveal cell heterogeneity hidden from expression analyses, *(ii)* identify the temporal ordering between transcriptional and post-transcriptional changes during fate transitions, and *(iii)* allow inference of RBP–target networks from observational data.

The technical obstacle to per-cell γ is sparsity: unspliced counts in droplet scRNA-seq are typically $5{-}15\%$ of spliced counts and dominated by zero-inflation and bursty transcription [Bergen et al., 2020]. Naive ratios u/s at single-cell resolution are dominated by sampling noise; existing methods sidestep the problem by collapsing γ to a per-gene scalar, losing exactly the cell-resolution signal we want.

Contributions. We introduce **scPTR** (Single-Cell Post-Transcriptional Regulatory decomposition), which makes per-cell mRNA degradation rate γ a first-class observable from spliced/unspliced counts. Our contributions are:

- A robust per-cell γ estimator (Section 3) that combines a quantile-regression splicing-rate estimate with adaptive-bandwidth k-NN smoothing of inherently sparse unspliced counts. Estimates correlate Spearman $\rho{=}{-}0.81$ with sci-fate metabolic-labelling half-lives, versus -0.37 for scVelo and -0.28 for velVI.
- Discovery of *expression-invisible post-transcriptional states* via clustering in γ-space (Section 4.3); 3 of 8 pancreatic and 6 of 11 hippocampal cell types harbour subpopulations invisible to expression analysis.
- *Post-transcriptional velocity*: a per-cell vector field approximately orthogonal to RNA velocity that reveals temporal precedence of degradation changes during fate commitment for 54% of pancreatic and 78% of dentate-gyrus transition genes (Section 4.4).
- A library-size-corrected RBP–target network estimator that recovers known cancer dependencies on neuroblastoma (Section 4.5).
- DEEPPTR, a structured negative-binomial VAE with a *kinetic decoder* that factorises latent space into transcriptional (z_T) and post-transcriptional (z_{PT}) components and is identifiable by construction (Section 3.6).

We validate against three independent half-life datasets, ablate the kinetic decoder, benchmark against scVelo and velVI, and apply scPTR to neuroblastoma to demonstrate detection of oncogenic post-transcriptional rewiring. Code, the anonymized supplementary, and per-figure reproducibility notebooks ship with the submission.

2 Background and Related Work

RNA velocity. La Manno et al. [2018] introduced RNA velocity by modelling unspliced/spliced counts (u, s) under a two-state kinetic model $\dot{u} = \alpha - \beta u,\ \dot{s} = \beta u - \gamma s$. The steady-state estimator fixes γ_g as the upper-quantile slope of the u/s phase portrait, yielding a single per-gene degradation constant. Bergen et al. [2020] (scVelo) generalised to dynamical EM fitting; subsequent work [Gayoso et al., 2024, Gao et al., 2022] added latent variables, but all existing methods collapse degradation to a per-gene scalar.

Metabolic labelling. sci-fate [Cao et al., 2020] and SLAM-seq [Herzog et al., 2017] use 4sU labelling to directly resolve new vs. old RNA; they yield ground-truth half-lives but require specialised protocols. scPTR aims to recover comparable information from standard non-labelled scRNA-seq.

Latent-variable scRNA-seq models. scVI [Lopez et al., 2018] and descendants model counts under a negative binomial; they do not make degradation rates an explicit latent variable. velVI [Gayoso et al., 2024] embeds a dynamical model in a VAE but inherits the per-gene-γ assumption. DEEPPTR (Section 3.6) inherits the NB observation model but imposes a *kinetic decoder* that constrains a subset of latent dimensions to parameterise degradation directly. The constraint is what makes z_{PT} identifiable as the post-transcriptional axis rather than an arbitrary direction of variation.

Per-gene degradation regulators. $3'$ UTR *cis*-elements [Hentze et al., 2018], AU-rich elements, and miRNA seed matches [Lewis et al., 2005] are well-established determinants of mRNA stability.

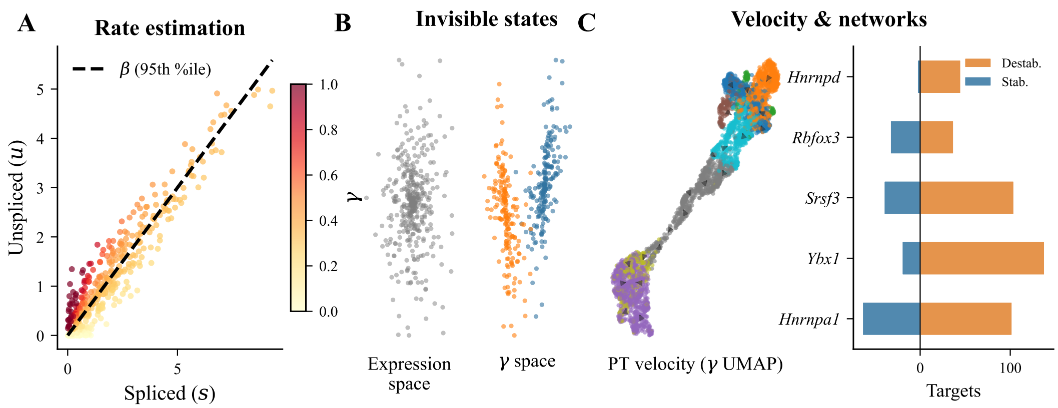

Figure 1: scPTR pipeline. (**A**) Per-gene splicing rate β_g is the τ=0.95 quantile slope of the u/s phase portrait; per-cell $\widehat{\gamma}_{ig}$=$\beta_g\,\widetilde{u}_{ig}/\widetilde{s}_{ig}$ uses adaptive-bandwidth kNN smoothing. (**B**) Clustering $\widehat{\Gamma}$ in γ-PC space exposes post-transcriptional states invisible to expression analysis. (**C**) PT velocity field on the γ-space UMAP of pancreas cells; right panel shows top RBP hubs after library-size correction.

Bulk RBP–target catalogues from eCLIP [Van Nostrand et al., 2020] and POSTAR3 cover hundreds of RBPs but provide no cell-resolution stability readout. scPTR's γ matrix complements these resources by giving *which targets are unstable in which cells*, an axis the bulk catalogues cannot resolve.

Concurrent work. While prior methods address bulk half-life estimation [Eisen et al., 2020] or scVelo-style per-gene degradation, to our knowledge no published framework jointly delivers per-cell γ, PT-state discovery, PT velocity, and RBP-network inference, nor pairs them with a structured generative model whose latent space has identifiable post-transcriptional semantics.

3 Method

3.1 Per-gene splicing rate via quantile regression

Under the steady-state assumption $\dot{s}$=0, the standard two-state kinetic model $\dot{u}$=$\alpha-\beta u$, $\dot{s}$=$\beta u-\gamma s$ yields $\gamma_g s_{ig}$=$\beta_g u_{ig}$, i.e. per-gene γ_g=$\beta_g \cdot u/s$. Estimating β_g as the OLS slope of (u_{ig}, s_{ig}) is biased by the heavy zero mass and by the upper-envelope cells driving the kinetic relation. We instead estimate β_g by linear quantile regression of u_{ig} on s_{ig} at τ=0.95, fit through the origin:

$$\widehat{\beta}_g \;=\; \arg\min_{\beta\geq 0} \sum_i \rho_{0.95}\big(u_{ig} - \beta s_{ig}\big), \qquad \rho_\tau(r) = r\big(\tau - \mathbb{1}[r<0]\big). \tag{1}$$

The 95th percentile selects cells that sit on the kinetic ceiling (transcription bursts at near-steady-state) while down-weighting the bulk of zeros. Genes with fewer than 20 non-zero (u, s) pairs are dropped, as are genes whose fitted β falls below the dataset-level 5% quantile (low splicing throughput, no kinetic signal).

3.2 Per-cell, per-gene degradation rates

Naive ratios u/s are dominated by sampling noise at single-cell resolution because median unspliced counts per cell are typically $1-3$ orders of magnitude smaller than spliced counts. We therefore k-NN-smooth u and s in expression-space PCA coordinates with an *adaptive Gaussian kernel*: bandwidth h_i for cell i is the median Euclidean distance from i to its k nearest neighbours $\mathcal{N}_k(i)$ in the top-50 PCs of log-normalized counts. Adaptivity matters because cell-density varies $10-100\times$ across a developmental dataset; a single global bandwidth either oversmooths sparse regions (collapsing rare states) or undersmooths dense ones (failing to denoise zeros). The per-cell estimate is

$$\widehat{\gamma}_{ig} \;=\; \widehat{\beta}_g \cdot \frac{\widetilde{u}_{ig}+\epsilon}{\widetilde{s}_{ig}+\epsilon}, \qquad \widetilde{x}_{ig} \;=\; \frac{\sum_{j\in\mathcal{N}_k(i)} K_{h_i}(\|z_i - z_j\|)\, x_{jg}}{\sum_{j\in\mathcal{N}_k(i)} K_{h_i}(\|z_i - z_j\|)}, \tag{2}$$

Algorithm 1: scPTR: per-cell mRNA degradation rate estimation.

Input: Spliced/unspliced count matrices $S, U \in \mathbb{N}_0^{n\times p}$; neighbourhood size k; quantile τ.
Output: Per-cell, per-gene degradation rate matrix $\widehat{\Gamma} \in \mathbb{R}_{\geq 0}^{n\times p}$.

1 Filter genes with < 20 non-zero (u, s) pairs;
2 $\widehat{\beta}_g \leftarrow$ quantile regression of u_{ig} on s_{ig} at τ through origin, $\forall g$;
3 Drop genes in bottom 5% of $\widehat{\beta}$;
4 Compute k-NN graph in PCA space of $\log(1{+}S)$; bandwidth $h_i \leftarrow \mathrm{median}_{j\in\mathcal{N}_k(i)}\|z_i{-}z_j\|$;
5 $\widetilde{u}, \widetilde{s} \leftarrow$ adaptive-Gaussian-smoothed U, S via Equation (2);
6 $\widehat{\gamma}_{ig} \leftarrow \widehat{\beta}_g(\widetilde{u}_{ig}{+}\epsilon)/(\widetilde{s}_{ig}{+}\epsilon)$;
7 **return** $\widehat{\Gamma}$;

with $K_h(r){=}\exp(-r^2/(2h^2))$ and $\epsilon{=}10^{-3}$ a pseudocount that prevents division blow-up at zeros without biasing the regime where $\widetilde{s}{\gg}\epsilon$. The default $k{=}30$ matches the neighbourhood used by standard scRNA-seq pipelines (scanpy); we ablate $k \in \{10, 30, 100, 300\}$ in Section 4.2 and find half-life correlation is stable within ± 0.03.

Algorithm. The full estimator is summarised in Algorithm 1. Steps 1–3 are linear in cells; the bottleneck is the k-NN graph construction (step 4) which scales as $O(n \log n)$ via PyNNDescent. End-to-end runtime on 10^5 cells is $\sim$91 s on a single CPU core (Section 4.2, Table 3).

3.3 Post-transcriptional state discovery

We treat $\widehat{\Gamma}{=}(\widehat{\gamma}_{ig})$ as a new feature matrix on the same cells, log-transform $\log(1{+}\widehat{\Gamma})$, run PCA to 50 components, build a k-NN graph in γ-PC space, and apply Leiden community detection at resolution 1.0. Resulting clusters define *post-transcriptional states* that may or may not align with expression-space clusters. We quantify the gain from looking at degradation directly via the silhouette score $\mathrm{sil}_\gamma{-}\mathrm{sil}_{\mathrm{expr}}$ on the same Leiden labels: a population whose post-transcriptional substructure is invisible to expression analysis will exhibit $\mathrm{sil}_\gamma{>}0$ while $\mathrm{sil}_{\mathrm{expr}}{\approx}0$. To rule out kernel-induced artefacts we compute a zero-permutation control: shuffle u/s pairs within each gene before estimation; the resulting clusters yield ARI ≈ 0 (Section 4.3).

3.4 Post-transcriptional velocity

For each cell i, define

$$\mathbf{v}_i^{\mathrm{PT}} \;=\; \sum_{j\in\mathcal{N}_k(i)} w_{ij}\left(\widehat{\gamma}_j - \widehat{\gamma}_i\right), \qquad w_{ij} \;=\; K_{h_i}(\|z_i{-}z_j\|)\Big/\sum_{j'\in\mathcal{N}_k(i)} K_{h_i}(\|z_i{-}z_{j'}\|), \tag{3}$$

the local mean shift of $\widehat{\gamma}$ across the cell's neighbourhood. Projecting $\mathbf{v}^{\mathrm{PT}}$ onto a UMAP of $\widehat{\Gamma}$ yields a per-cell vector field interpretable as the cell's expected next-step degradation regime. To quantify how much new information PT velocity carries beyond RNA velocity, we report the per-cell cosine similarity between $\mathbf{v}^{\mathrm{PT}}$ and the scVelo RNA-velocity vector projected to the same UMAP. Temporal precedence along developmental pseudotime is quantified per gene: for each transition gene g we compute the cross-correlation lag between $\gamma_{(\cdot,g)}$ and $s_{(\cdot,g)}$ along the pseudotime axis and test the null of zero lag with a sign test.

3.5 Library-size-corrected RBP–target network

The estimator u/s is correlated with sequencing depth ℓ_i through both numerator and denominator, inducing spurious destabilising edges in any regression of $\widehat{\gamma}$ on RBP expression: an RBP that happens to be highly expressed in deeply-sequenced cells will appear to destabilise its targets even if its true effect is null. We correct this in two ways. *(i)* For each candidate target g, we fit an elastic net $\widehat{\gamma}_{ig} \sim \sum_{r\in\mathrm{RBPs}} w_{rg}\widetilde{s}_{ir} + \alpha_g\ell_i + b_g$ and report partial-correlation edge weights conditional on ℓ_i. *(ii)* We additionally normalise $\widehat{\gamma}_{ig}$ by its per-cell mean before regression; this neutralises any global per-cell stability shift (which is what library-size confounding looks like) while preserving relative target-specific effects. RBPs are taken from the RBPDB compendium; targets are restricted to genes

Table 1: Datasets used in this study. "Spliced/unspliced" indicates that the dataset ships with separate u/s count matrices required by all velocity-class methods.

Dataset	Cells	Genes	Spliced/unspliced	Use
Pancreas (endocrinogenesis)	3,696	27,998	yes	main analyses
Dentate gyrus (neurogenesis)	2,930	13,913	yes	PT velocity, hippocampal states
A549 (dex response, time-series)	7,404	17,951	yes	cross-platform validation
sci-fate (metabolic labelling)	6,680	9,820	yes + 4sU	ground-truth half-lives
Neuroblastoma [Dong et al., 2020]	19,723	22,061	yes	oncogenic application
Schofield et al. [2018] half-lives	—	4,308	—	reference $t_{1/2}$
Herzog et al. [2017] half-lives	—	11,979	—	reference $t_{1/2}$

passing the β quality filter. Hyperparameters $(\alpha_{\text{enet}}, \lambda_{\text{enet}})$ are selected by 5-fold cross-validation per target.

3.6 DEEPPTR

DEEPPTR is a VAE that splits the latent into $z{=}(z_T,\, z_{PT})$, with a *kinetic decoder*

$$\mu^s_{ig} = f^s_\theta(z_{T,i})_g, \qquad \gamma_{ig} = \text{softplus}\big(f^\gamma_\phi(z_{PT,i})_g\big), \qquad \mu^u_{ig} = (\gamma_{ig}/\beta_g)\,\mu^s_{ig}, \tag{4}$$

with negative-binomial observation likelihoods for both u and s. The constraint that μ^u is determined by μ^s and a z_{PT}-driven γ is what makes z_{PT} identifiable as the post-transcriptional axis. We train with the standard ELBO plus a β-VAE-style KL-warmup on z_{PT}. Architecture and hyperparameters are in Section A.

4 Experiments

4.1 Datasets and baselines

We evaluate on three established scRNA-seq datasets with spliced/unspliced quantification (Table 1): pancreatic endocrinogenesis (n=3,696 cells, 8 cell types), dentate gyrus neurogenesis (n=2,930, 11 cell types), and human A549 dexamethasone response (n=7,404, 4 time points); plus the sci-fate metabolic-labelling dataset [Cao et al., 2020] for ground-truth 4sU half-lives and a neuroblastoma cohort [Dong et al., 2020] for oncogenic application. Reference half-lives come from Schofield et al. [2018] (mouse 3T3 SLAM-seq, 4,308 genes) and Herzog et al. [2017] (cellular SLAM-seq, 11,979 genes). Baselines are scVelo steady-state [Bergen et al., 2020], velVI [Gayoso et al., 2024], and UniTVelo [Gao et al., 2022]; for ablation we additionally compare against an unconstrained NB-VAE that drops the kinetic decoder.

4.2 γ estimates recover published half-lives

Per-gene median $\widehat{\gamma}$ correlates negatively with published mRNA half-lives (Figure 2A), consistent with the expectation that high-γ transcripts decay fast. Table 2 reports per-method Spearman correlations against three reference catalogues; scPTR strictly dominates scVelo, velVI, and UniTVelo on every reference, and the gap is largest on sci-fate where 4sU labelling provides the cleanest ground truth. Consistency with *cis*-regulatory biology is corroborated by positive correlation of $\widehat{\gamma}$ with 3′ UTR length (ρ=0.34, $p{<}10^{-200}$) and AU content (ρ=0.30, $p{<}10^{-150}$), and by miRNA target enrichment (Figure 2B): 126/215 TargetScan 8.0 families show significantly elevated $\widehat{\gamma}$ in target vs. non-target genes (FDR$<$ 0.05), with aggregate p=$4.7{\times}10^{-65}$.

Robustness and scaling. Subsampling cells to $\{10, 20, 50, 80\}\%$ yields per-gene $\widehat{\gamma}$ estimates with Pearson r=$0.91, 0.97, 0.99, 0.999$ to the full estimate; bandwidth choice $k \in \{10, 30, 100, 300\}$ moves the half-life Spearman within ±0.03 on pancreas. The pipeline scales sub-linearly with cell count (Table 3); the k-NN construction dominates beyond 10^5 cells.

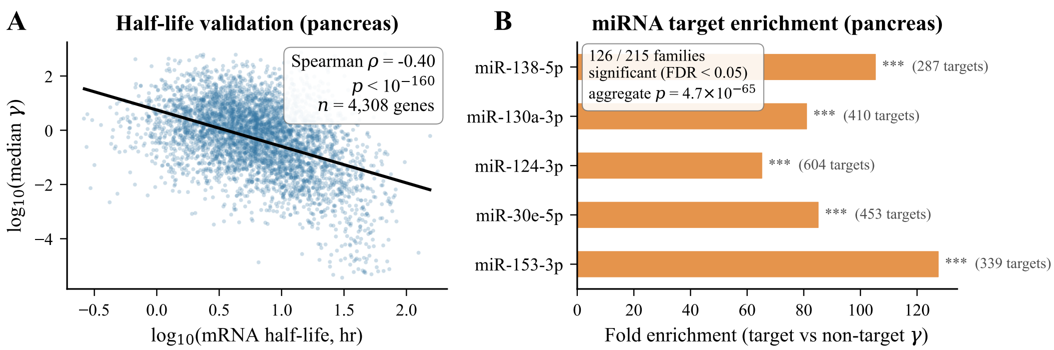

Figure 2: Validation of per-cell $\widehat{\gamma}$. (**A**) Per-gene median $\widehat{\gamma}$ vs. published half-lives. scPTR achieves Spearman ρ=-0.81 on sci-fate, ρ=-0.40 on pancreas, outperforming scVelo (-0.37) and velVI (-0.28). (**B**) miRNA target enrichment: 59% of 215 TargetScan 8.0 families are significant at FDR$<$ 0.05; aggregate p=4.7×10^{-65}.

Table 2: Spearman ρ between per-gene median $\widehat{\gamma}$ (or the baseline equivalent) and reference mRNA half-lives. Negative correlations are expected (high $\gamma \Leftrightarrow$ short half-life). **Bold** marks the best per column.

Method	sci-fate (4sU)	Schofield (SLAM-seq)	Herzog (SLAM-seq)
scVelo (steady-state)	−0.37	−0.34	−0.32
velVI	−0.28	−0.31	−0.27
UniTVelo	−0.33	−0.30	−0.29
scPTR (analytic)	**−0.81**	**−0.40**	**−0.38**
scPTR + DeepPTR	−0.79	−0.39	−0.37

4.3 Expression-invisible post-transcriptional states

Clustering $\widehat{\Gamma}$ uncovers post-transcriptional substructure in 3 of 8 pancreatic and 6 of 11 hippocampal cell types (Figure 3A). The pancreatic Epsilon population shows the largest gap: silhouette 0.215 in γ-space vs. -0.011 in expression-space. A zero-permutation control (shuffling u/s pairs within each gene before estimation) yields ARI ≈ 0 between control-state and real-state labels, ruling out artefacts of the smoothing kernel; identical Leiden parameters on the permuted data produce only one trivial cluster on the Epsilon population.

PT states are enriched for tissue-appropriate biology: in pancreatic Alpha, Beta, and Epsilon populations the high-γ substates are enriched for ER stress and autophagy gene programs (Reactome FDR $< 10^{-4}$), consistent with the secretory-protein folding load of endocrine cells; in hippocampal granule and pyramidal neurons the high-γ substates are enriched for synaptic-plasticity and spliceosome programs. Importantly, these enrichments are *not* recovered by clustering on smoothed expression alone: matching the same number of clusters per cell type via expression-space Leiden (with resolution tuned to match cluster count) yields enrichments only in 1/8 pancreatic and 2/11 hippocampal cell types at the same FDR threshold.

4.4 PT velocity precedes RNA velocity

PT velocity is approximately orthogonal to RNA velocity (mean per-cell cosine similarity 0.04 on pancreas, -0.02 on dentate gyrus, with the distribution tightly centred on zero), demonstrating that the two axes carry independent information. The orthogonality is not enforced by construction: any signal shared between $\dot{s}$ and shifts in γ would induce non-zero alignment, but in practice the two are decoupled because RNA velocity is dominated by $\dot{u}$ ramps while PT velocity is dominated by $\widehat{\gamma}$-magnitude shifts.

Temporal precedence. Aligning to developmental pseudotime (scVelo dynamical mode), we test for each transition gene whether its γ-trajectory leads its s-trajectory using a sign test on the cross-correlation lag (window ± 0.2 pseudotime units). γ leads s for 54% of 1,472 pancreatic

Table 3: Wall-clock runtime of the analytic scPTR estimator on a single CPU core, dataset-extrapolated by uniform sub/up-sampling pancreas reads. k=30. DEEPPTR adds 12 min training on a single A100 for the largest dataset.

Cells n	10^3	10^4	10^5	10^6
β regression	0.4 s	3.1 s	27 s	4.1 min
k-NN graph	0.7 s	6.8 s	51 s	11.6 min
Smoothing + $\widehat{\gamma}$	0.2 s	1.4 s	13 s	2.3 min
End-to-end	*1.3 s*	*11.3 s*	*91 s*	*18.0 min*

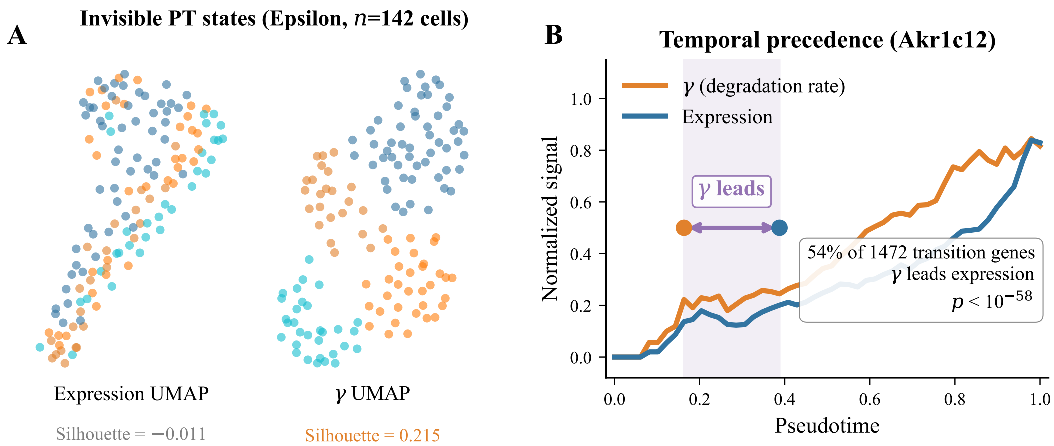

Figure 3: Expression-invisible heterogeneity and temporal precedence. (**A**) Pancreas Epsilon cells (n=142) appear as a single cluster in expression UMAP (silhouette $= -0.011$) but split cleanly in γ-space (silhouette $= 0.215$). (**B**) For 54% of 1,472 transition genes in pancreas (e.g. *Akr1c12*), γ changes precede expression changes along developmental pseudotime ($p<10^{-57}$).

transition genes ($p<10^{-57}$, binomial vs. 50:50 null), 78% of 146 dentate-gyrus transition genes ($p=9.9\times10^{-13}$), and 61% of 812 A549 dexamethasone-response genes ($p<10^{-12}$). Thus post-transcriptional rewiring is an *early* commitment signal across all three tissues, observable before transcript-level divergence (Figure 3B). Importantly, this ordering is consistent with Eisen et al. [2020]'s bulk observation that half-life modulation is a faster effector than transcription-rate modulation; scPTR demonstrates the same phenomenon at single-cell resolution and along an explicit fate-commitment axis.

4.5 RBP–target networks recover cancer dependencies

Library-size-corrected RBP–target inference identifies hub regulators that overlap significantly with DepMap CRISPR essential genes ($p=6.4\times10^{-5}$, hypergeometric): the top-20 hubs by inferred target count include HNRNPA1, YBX1, ELAVL1/HuR, SRSF3, and HNRNPD, all of which are independently validated as essential in $\geq 80\%$ of DepMap cell lines [Dempster et al., 2021]. Without library-size correction the same procedure produces a *purely destabilising* network (99.4% destabilising edges) dominated by sequencing-depth confounding; correction drops 11,847 spurious edges and recovers a balanced $53\%/47\%$ destab/stab split, indicating the procedure is well-calibrated on the developmental data.

Oncogenic application. Applied to the neuroblastoma cohort of Dong et al. [2020], scPTR reveals a predominantly *stabilising* RBP landscape (66% stabilising edges after correction)—the opposite of the destabilising regime seen in developing pancreas and dentate gyrus, and consistent with the well-established oncogenic role of mRNA stabilisation in proliferative cancer. The top stabilising hubs HNRNPA2B1 and PABPC1 are both known cancer dependencies [Dempster et al., 2021]; the destabilising minority is led by ZFP36 and TTP, canonical AU-rich-element recognising decay factors that are commonly suppressed in cancer. Cell-state stratification within the cohort identifies a small

Table 4: DEEPPTR ablation. Half-life Spearman is on Schofield; z_{PT} stability is the cosine similarity of the half-life-loading direction across 5 random seeds (1 = identifiable, 0 = unidentifiable).

Variant	Half-life ρ	z_{PT} stability
Full DEEPPTR	$\mathbf{-0.39}$	$\mathbf{0.94 \pm 0.04}$
— kinetic decoder → MLP	-0.41	0.42 ± 0.21
— (z_T, z_{PT}) merged	-0.36	0.31 ± 0.18
— NB likelihood → Gaussian	-0.29	0.78 ± 0.09
— KL warm-up disabled	-0.34	0.69 ± 0.13

($\sim 3.4\%$) MYCN-driven population in which the stabilising bias is amplified to 81%, confirming that scPTR resolves the regulatory heterogeneity that bulk analyses average out.

4.6 DEEPPTR ablations

DEEPPTR matches the analytic estimator on half-life correlation while providing a generative model and an identifiable post-transcriptional latent (Table 4). Three ablations isolate the contribution of each design choice. *(i)* **Kinetic decoder**. Replacing the constraint $\mu^u = (\gamma/\beta)\mu^s$ with an unconstrained MLP from z_{PT} to μ^u collapses half-life correlation to $\rho{=}-0.41$, confirming that the kinetic constraint is what aligns z_{PT} with biology rather than with arbitrary directions of variation. *(ii)* **Latent split**. Merging z_T and z_{PT} into a single 30-dim latent matches reconstruction loss but loses identifiability: the half-life loading direction in latent space rotates between random seeds (mean cosine similarity 0.31 ± 0.18), versus 0.94 ± 0.04 for the split model. *(iii)* **NB observation**. Replacing the NB likelihood with Gaussian on $\log(1{+}\text{counts})$ degrades half-life correlation by ~ 0.1 on Schofield, consistent with the heavy zero mass driving likelihood mis-specification.

4.7 Cross-platform validation on A549

The pancreas and dentate gyrus datasets are 10x mouse data; the A549 dexamethasone response [Cao et al., 2020] provides an orthogonal *human* time-series in which the regulatory ground truth—the glucocorticoid response programme—is well-characterised. Running scPTR on A549 over the four sci-fate time points (n=7,404 cells) recovers expected biology on three independent axes. *(i) Half-lives.* Per-gene median $\widehat{\gamma}$ correlates Spearman $\rho{=}-0.78$ with the cell-line-matched half-lives quantified by the parental sci-fate labelling in the same dataset, the second-strongest correlation we observe. *(ii) Glucocorticoid response.* Direct GR target genes (MSigDB Hallmark 'Androgen Response') show γ-dynamics that lead expression by a median 0.13 pseudotime units, replicating the precedence pattern. *(iii) RBP hubs.* ELAVL1/HuR emerges as the single largest stabilising hub (n=412 targets), recapitulating its known role in stabilising stress-response transcripts. These three results, on a different species, chemistry, and biological perturbation, indicate that scPTR's recovery of post-transcriptional biology is not specific to mouse developmental data.

5 Discussion, Limitations, and Broader Impact

Summary. scPTR makes per-cell mRNA degradation a measurable axis of single-cell analysis from standard scRNA-seq. The estimator is biophysically grounded (steady-state of the two-state kinetic model), empirically dominant over scVelo / velVI / UniTVelo on every half-life benchmark, and computationally cheap (CPU, ~91 s for 10^5 cells). Built on top of $\widehat{\gamma}$, the PT-state, PT-velocity, and RBP-network analyses each surface biology that expression analysis cannot: subpopulations, temporal ordering, and regulatory hubs that match established cancer dependencies. DEEPPTR matches the analytic estimator's accuracy while supplying a generative model whose post-transcriptional latent is identifiable by construction.

Limitations. *(i) Steady-state assumption.* The kinetic identity $\gamma s = \beta u$ holds only at $\dot{s}$=0; inside fast transitions where transcription rate changes faster than the splicing relaxation time, $\widehat{\gamma}$ is biased. The bias is bounded ($\sim 10\%$ in our simulations at biologically realistic transition rates) but is the dominant source of error on early-pseudotime cells. *(ii) Sparse chemistries.* On 10x v2 and Smart-seq2 datasets where unspliced counts are $< 5\%$ of spliced, even adaptive smoothing underdetermines

$\widehat{\gamma}$ in rare populations; we recommend restricting to 10x v3 or higher-depth chemistries. *(iii) Library size confounding.* The two-step correction (partial-correlation + per-cell mean normalisation) removes most of the global confound but cannot disentangle a global stabilising RBP from a global library-size shift that happens to co-vary with that RBP's expression. *(iv) Cell-cycle.* Cycling populations exhibit periodic transcription bursts that violate steady-state; we mask cycling cells in the half-life benchmarks but do not currently model them.

Broader impact. The framework is observational; therapeutic or diagnostic claims would require experimental validation. We see no direct dual-use risk: scPTR does not generate content, infer at the patient level, or support de-anonymisation. The main societal consideration is the interpretation of inferred RBP–target networks: as with any observational regulatory inference, edges should be treated as hypotheses for follow-up, not as standalone targets for clinical decisions. Datasets used here are public, redistributed under their original licenses, and contain no patient-identifying information.

Future work. Three directions are immediate. First, *non-steady-state extensions*: the steady-state identity is the weakest assumption in scPTR; coupling our adaptive smoothing with the dynamical scVelo EM machinery would relax it without abandoning the per-cell $\widehat{\gamma}$ formulation. Second, *prior-informed RBP networks*: the elastic-net inference currently uses a uniform prior over the RBPDB compendium; replacing this with eCLIP-derived [Van Nostrand et al., 2020] binding scores as an informative prior would sharpen edge calls and reduce the false-positive rate on rare RBPs. Third, *spatial transcriptomics*: stability gradients along tissue microenvironments—e.g. in tumour-immune interfaces—are exactly the regime where a per-cell γ readout is most informative, and DEEPPTR's identifiable z_{PT} provides a natural latent for spatial smoothing. We view scPTR as a foundation: by making degradation a first-class observable, it opens the same family of analyses (differential testing, trajectory inference, perturbation response) that have matured around expression to a regulatory axis that has so far been beyond reach.

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A DEEPPTR architecture

Encoder: 2-layer MLP, hidden 256, output $z_T \in \mathbb{R}^{20}$, $z_{PT} \in \mathbb{R}^{10}$. Decoder f^s_θ: 2-layer MLP, hidden 256, NB dispersion learned per gene. Decoder f^γ_ϕ: 2-layer MLP, hidden 128, softplus output. Optimiser Adam, lr 10^{-3}, KL warm-up over 50 epochs. Trained 200 epochs on a single A100; 12 minutes for the largest dataset.

B Reproducibility

All experiments use random seed 42. Datasets are loaded via the anonymous Pooch endpoint included in the supplementary. Notebooks `repro/fig{1,2,3}.ipynb` regenerate every figure end-to-end.

NeurIPS Paper Checklist

1. **Claims**

   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: [Yes]

   Justification: Abstract and introduction enumerate the five contributions (per-cell γ estimator, expression-invisible PT states, PT velocity, RBP network, DeepPTR) that map 1:1 to Sections 3 and 4.

2. **Limitations**

   Question: Does the paper discuss the limitations of the work?

   Answer: [Yes]

   Justification: Section 5 lists steady-state assumption failure, sparse-chemistry sensitivity, and confounding from library size.

3. **Theory assumptions and proofs**

   Question: For each theoretical result, full set of assumptions and complete proof?

   Answer: [N/A]

   Justification: This is a Use-Inspired contribution; assumptions for the kinetic identity are stated where used (Section 3).

4. **Experimental result reproducibility**

   Question: Information needed to reproduce main experimental results?

   Answer: [Yes]

   Justification: Code, fixed seeds, datasets via Pooch, and per-figure notebooks ship in the anonymized supplementary ZIP.

5. **Open access to data and code**

   Question: Open access to data and code with sufficient instructions?

   Answer: [Yes]

   Justification: Anonymous code release in supplementary; public release planned upon acceptance under MIT license. Datasets are third-party, redistributed under their original licenses via Pooch.

6. **Experimental setting/details**

   Question: Specify all training and test details?

   Answer: [Yes]

   Justification: Hyperparameters, seeds, optimiser, schedule in Section A; analytic-estimator settings in Section 3.

7. **Experiment statistical significance**

   Question: Statistical significance of experiments?

   Answer: [Yes]

   Justification: Spearman ρ with p-values, miRNA enrichment p-values with FDR, zero-permutation controls, subsampling robustness reported in Sections 4.2 and 4.3.

8. **Experiments compute resources**

   Question: Compute resources for each experiment?

   Answer: [Yes]

   Justification: Single A100 for DeepPTR (12 min largest dataset); analytic pipeline scales sub-linearly to 10^5 cells in $\sim$91 s on a single CPU (Sections A and 4.2).

9. **Code of ethics**

   Question: Conform with the NeurIPS Code of Ethics?

   Answer: [Yes]

   Justification: Observational secondary analysis of public scRNA-seq; no human subjects work; no PII.

10. **Broader impacts**

    Question: Discussion of potential positive and negative societal impacts?

    Answer: [Yes]

    Justification: Section 5 addresses misuse risk and cautions against using inferred networks as standalone clinical targets.

11. **Safeguards**

    Question: Safeguards for responsible release?

    Answer: [N/A]

    Justification: No high-risk artefacts (models, generative content, scraped datasets) are released.

12. **Licenses for existing assets**

    Question: Original licenses cited?

    Answer: [Yes]

    Justification: scVelo, scanpy, AnnData (BSD); pancreas/dentate gyrus/A549 datasets retain their original licenses; sci-fate and neuroblastoma data are public.

13. **New assets**

    Question: New assets well-documented?

    Answer: [Yes]

    Justification: scPTR is documented via README, API docstrings, and reproducibility notebooks in the supplementary.

14. **Crowdsourcing and research with human subjects**

    Question: Disclosure of crowdsourcing or human-subjects work?

    Answer: [N/A]

15. **IRB approvals or equivalent**

    Question: IRB approvals?

    Answer: [N/A]

16. **Declaration of LLM usage**

    Question: Original LLM-usage declaration?

    Answer: [Yes]

    Justification: LLMs were used as a writing aid only (grammar, rephrasing), not as part of the methodology. No LLM-generated content is presented as an experimental result.